Experiment No. 02

AIM

Apply Data Imputation Techniques on the given dataset.

THEORY

What is Imputation?

Imputation is the process of replacing **missing data** (null, NaN, or NA values) with substituted values. Real-world datasets often contain missing entries due to data entry errors, sensor failures, survey non-responses, or system glitches. Imputation allows us to retain as much data as possible while still performing meaningful analysis.

Why is Imputation Important?

- Most machine learning algorithms **cannot handle missing values** directly.
- Dropping rows/columns with missing data can lead to **significant information loss**.
- Proper imputation can **reduce bias** and improve model performance.
- It helps maintain the **statistical power** of the dataset.

Data Imputation Techniques

#	Technique	Type	Description	Advantages	Disadvantages
1	**Deletion of rows**	Deletion	Remove rows containing missing values	Simple, no assumptions	Data loss, biased if data is not MCAR
2	**Mean Imputation**	Statistical	Replace missing numerical values with column mean	Easy, preserves mean	Reduces variance, ignores correlations
3	**Median Imputation**	Statistical	Replace missing numerical values with column median	Robust to outliers	Reduces variance, ignores relationships
4	**Mode Imputation**	Statistical	Replace missing categorical values with most frequent category	Simple for categorical data	Can overrepresent dominant category
5	**Arbitrary Value Imputation**	Value-based	Replace missing values with a chosen constant (e.g., 0, 999)	Flags missing data explicitly	Can distort distribution
6	**End of Tail**	Statisti	Replace with	Distinguishes	Creates outliers,

#	Technique	Type	Description	Advantages	Disadvantages
	Imputation	cal	values at the extreme end (mean ± 3*std)	imputed values	distorts distribution
7	**Random Sample Imputation**	Sampling	Fill missing values with random samples from non-missing data	Preserves original distribution	Adds randomness, not reproducible without seed
8	**Frequent Category Imputation**	Categorical	Replace missing categories with the most frequent one	Simple, intuitive	Overrepresents one category
9	**Adding "Missing" Category**	Categorical	Treat missing as a separate category called "Missing"	Retains missingness information	Increases cardinality
10	**Regression Imputation**	Model-based	Predict missing values using a regression model	Preserves relationships between features	Assumes linear relationship, overfits

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Dataset Used

Online Retail Dataset — Contains transactional data from a UK-based online retail store (Dec 2010 – Dec 2011).

Libraries Used

- `pandas`, `numpy` — Data manipulation
- `missingno` — Missing data visualization
- `seaborn`, `matplotlib` — Statistical plotting
- `plotly` — Interactive visualizations
- `scikit-learn` — Regression imputation

```
# Install required libraries
!python -m pip install missingno plotly -q

# Import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import missingno as msno
import plotly.express as px
import plotly.graph_objects as go
from sklearn.linear_model import LinearRegression
import warnings
warnings.filterwarnings('ignore')
```

```python
print("All libraries imported successfully!")
```

```
All libraries imported successfully!
```

```python
# Load the dataset
df = pd.read_csv('data_online_retail.csv', encoding='latin1')
print(f"Dataset Shape: {df.shape}")
df.head(10)
```

```
Dataset Shape: (541909, 8)
```


```
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{"name":"StockCode","rawType":"object","type":"string"},
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BABUSHKA NESTING BOXES","2","12/1/2010 8:26","7.65","17850.0","United
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WARMER RED POLKA DOT","6","12/1/2010 8:28","1.85","17850.0","United
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Kingdom"]],"shape":{"columns":8,"rows":10}}
```


```python
# Basic dataset info
print("Dataset Info:")
print(df.info())
print("\nDescriptive Statistics:")
df.describe(include='all')
```

```
Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 541909 entries, 0 to 541908
Data columns (total 8 columns):
```

```
 #   Column       Non-Null Count   Dtype
---  ------       --------------   -----
 0   InvoiceNo    541909 non-null  object
 1   StockCode    541909 non-null  object
 2   Description  540455 non-null  object
 3   Quantity     541909 non-null  int64
 4   InvoiceDate  541909 non-null  object
 5   UnitPrice    541909 non-null  float64
 6   CustomerID   406829 non-null  float64
 7   Country      541909 non-null  object
dtypes: float64(2), int64(1), object(5)
memory usage: 33.1+ MB
None

Descriptive Statistics:
```


```
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```


1. Analyzing Missing Data

Before imputation, let's understand the extent and pattern of missing values in the dataset.

```
# Check missing values
missing = df.isnull().sum()
```

```
missing_pct = (df.isnull().sum() / len(df)) * 100
missing_df = pd.DataFrame({'Missing Count': missing, 'Missing %':
missing_pct})
missing_df = missing_df[missing_df['Missing Count'] >
0].sort_values('Missing %', ascending=False)
print("Missing Values Summary:")
print(missing_df)
print(f"\nTotal rows: {len(df)}")
print(f"Rows with at least one missing value:
{df.isnull().any(axis=1).sum()}")
```

```
Missing Values Summary:
             Missing Count  Missing %
CustomerID          135080  24.926694
Description           1454   0.268311

Total rows: 541909
Rows with at least one missing value: 135080
```

```
# Introduce additional missing values for demonstration purposes
np.random.seed(42)
df_demo = df.copy()

# Add missing values to 'Description' (categorical), 'Quantity'
(numerical), 'UnitPrice' (numerical)
mask_desc = np.random.choice(df_demo.index, size=int(len(df_demo) *
0.05), replace=False)
mask_qty = np.random.choice(df_demo.index, size=int(len(df_demo) *
0.08), replace=False)
mask_price = np.random.choice(df_demo.index, size=int(len(df_demo) *
0.06), replace=False)

df_demo.loc[mask_desc, 'Description'] = np.nan
df_demo.loc[mask_qty, 'Quantity'] = np.nan
df_demo.loc[mask_price, 'UnitPrice'] = np.nan

# Convert Quantity and UnitPrice to float to hold NaN
df_demo['Quantity'] = df_demo['Quantity'].astype(float)
df_demo['UnitPrice'] = df_demo['UnitPrice'].astype(float)

print("Missing values after introducing NaNs for demonstration:")
print(df_demo.isnull().sum())
print(f"\nDataset shape: {df_demo.shape}")
```

```
Missing values after introducing NaNs for demonstration:
InvoiceNo           0
StockCode           0
Description     28480
Quantity        43352
InvoiceDate         0
```

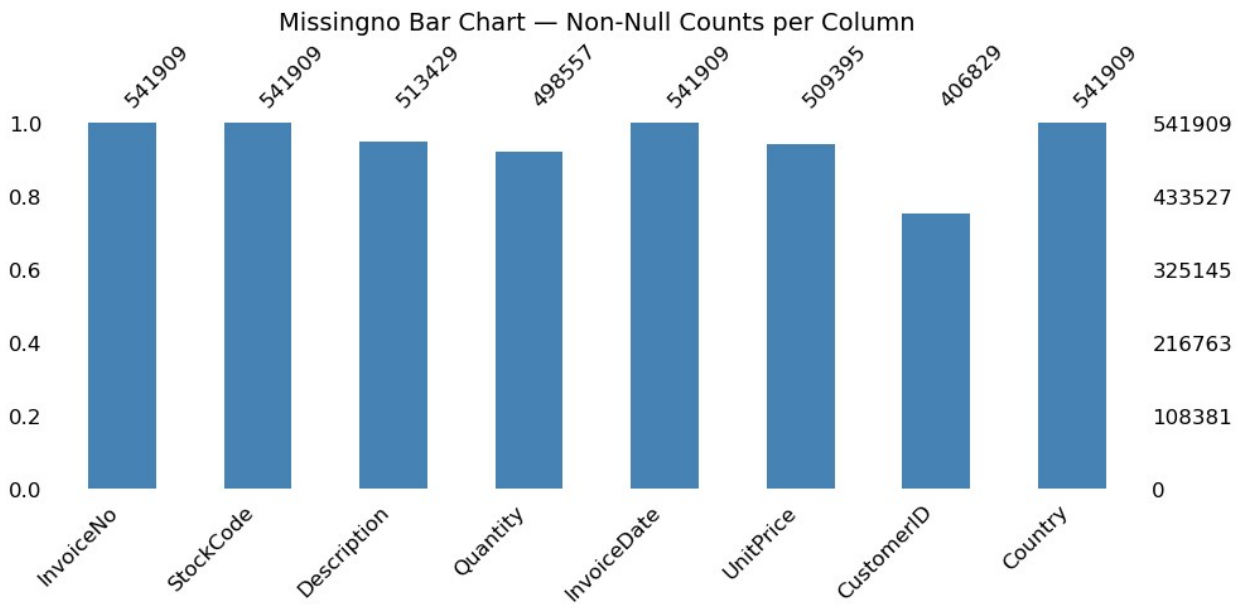
```
UnitPrice       32514
CustomerID     135080
Country             0
dtype: int64

Dataset shape: (541909, 8)
```

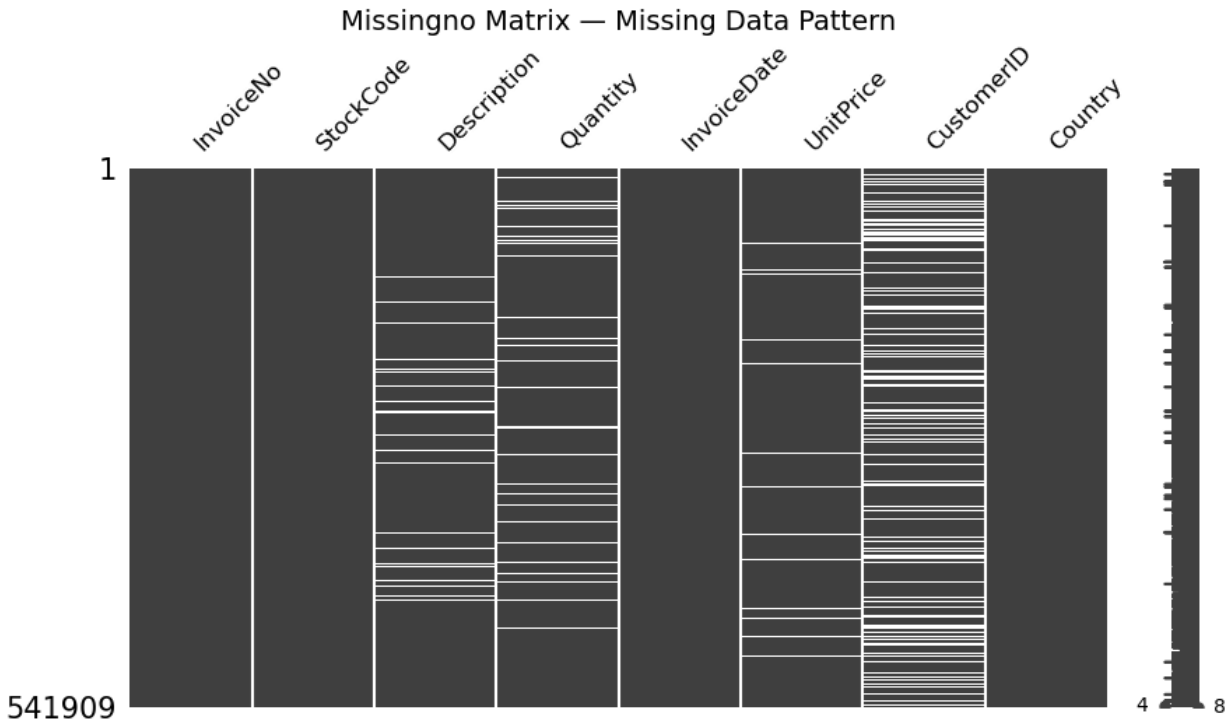
2. Visualizing Missing Data

Using **missingno**, **seaborn**, and **plotly** to visualize the pattern of missingness.

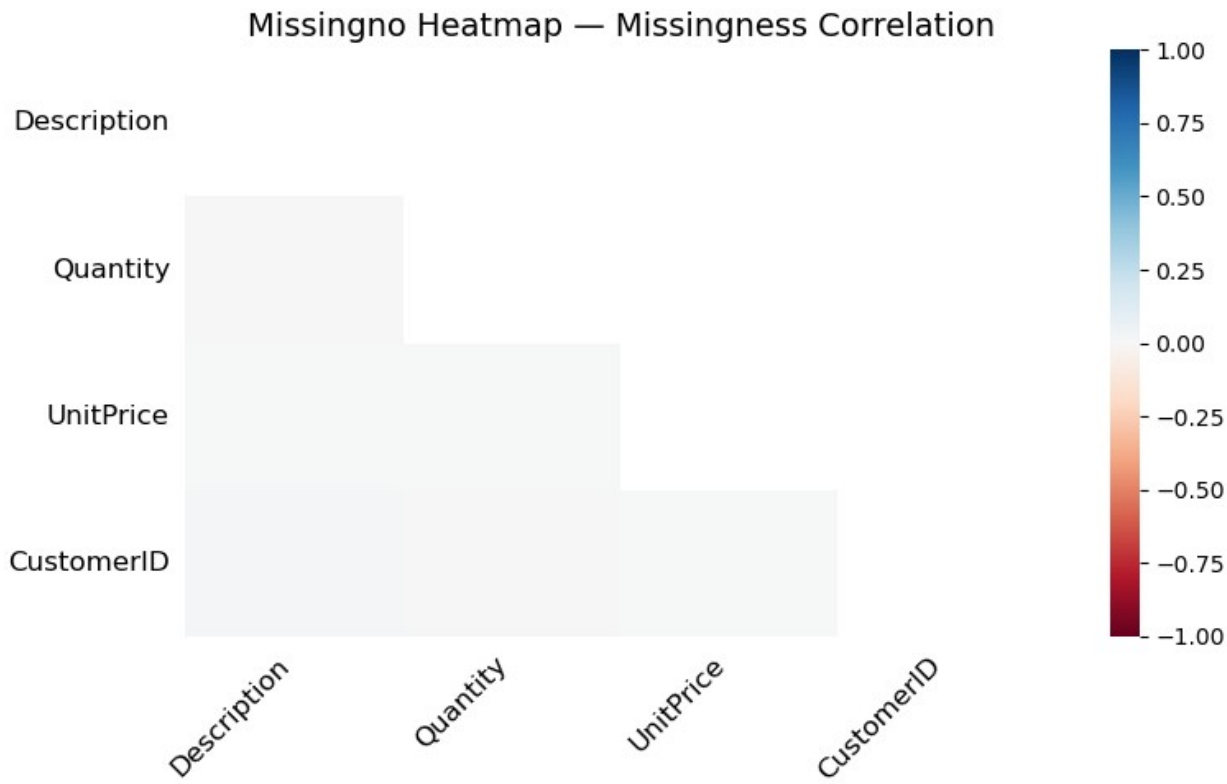
```
# Missingno Bar Chart — shows non-null counts per column
msno.bar(df_demo, figsize=(10, 5), color='steelblue', fontsize=12)
plt.title("Missingno Bar Chart — Non-Null Counts per Column",
fontsize=14)
plt.tight_layout()
plt.show()
```



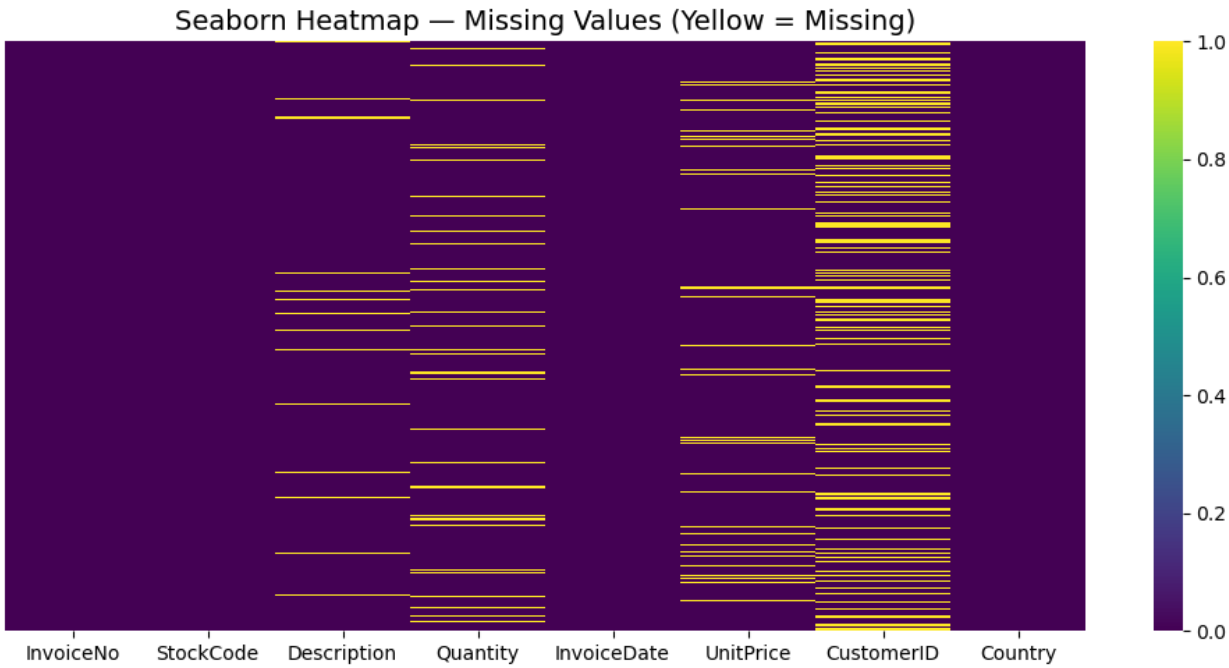

```
# Missingno Matrix — visualize missing value locations
msno.matrix(df_demo, figsize=(10, 5), fontsize=12)
plt.title("Missingno Matrix — Missing Data Pattern", fontsize=14)
plt.tight_layout()
plt.show()
```




```python
# Missingno Heatmap — correlation of missingness between columns
msno.heatmap(df_demo, figsize=(8, 5), fontsize=12)
plt.title("Missingno Heatmap — Missingness Correlation", fontsize=14)
plt.tight_layout()
plt.show()
```




```
# Seaborn Heatmap – missing values heatmap (True/False)
plt.figure(figsize=(10, 5))
sns.heatmap(df_demo.isnull(), cbar=True, yticklabels=False,
cmap='viridis')
plt.title("Seaborn Heatmap – Missing Values (Yellow = Missing)",
fontsize=14)
plt.tight_layout()
plt.show()
```




```
# Plotly Bar Chart — Missing values percentage per column
missing_pct_demo = (df_demo.isnull().sum() / len(df_demo)) * 100
fig = px.bar(
    x=missing_pct_demo.index,
    y=missing_pct_demo.values,
    labels={'x': 'Column', 'y': 'Missing %'},
    title='Missing Values Percentage per Column (Plotly)',
    color=missing_pct_demo.values,
    color_continuous_scale='Reds'
)
fig.update_layout(xaxis_tickangle=-45)
fig.show()
```

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```


3. Data Imputation Techniques

Technique 1: Deletion of Rows with Missing Data

Remove all rows that contain at least one missing value.

```
# Technique 1: Deletion of Rows with Missing Data
df_deleted = df_demo.dropna()
print(f"Original shape: {df_demo.shape}")
print(f"After row deletion: {df_deleted.shape}")
print(f"Rows removed: {len(df_demo) - len(df_deleted)}
({((len(df_demo) - len(df_deleted)) / len(df_demo)) * 100:.2f}%)")
```

```
print(f"\nMissing values after deletion:\
n{df_deleted.isnull().sum()}")
```

```
Original shape: (541909, 8)
After row deletion: (334230, 8)
Rows removed: 207679 (38.32%)

Missing values after deletion:
InvoiceNo      0
StockCode      0
Description    0
Quantity       0
InvoiceDate    0
UnitPrice      0
CustomerID     0
Country        0
dtype: int64
```

Technique 2: Mean Imputation

Replace missing numerical values with the **mean** of the column.

```
# Technique 2: Mean Imputation
df_mean = df_demo.copy()

mean_qty = df_mean['Quantity'].mean()
mean_price = df_mean['UnitPrice'].mean()

df_mean['Quantity'].fillna(mean_qty, inplace=True)
df_mean['UnitPrice'].fillna(mean_price, inplace=True)

print(f"Mean of Quantity: {mean_qty:.2f}")
print(f"Mean of UnitPrice: {mean_price:.2f}")
print(f"\nMissing values after Mean Imputation:")
print(df_mean[['Quantity', 'UnitPrice']].isnull().sum())

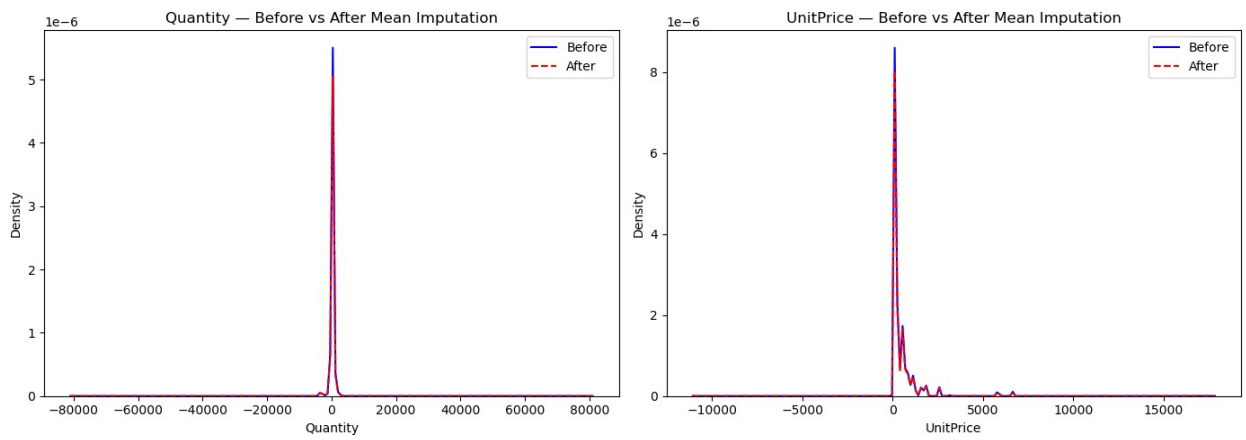
# Visualize the distribution before and after
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
axes[0].set_title("Quantity — Before vs After Mean Imputation")
sns.kdeplot(df_demo['Quantity'].dropna(), ax=axes[0], label='Before',
color='blue')
sns.kdeplot(df_mean['Quantity'], ax=axes[0], label='After',
color='red', linestyle='--')
axes[0].legend()

axes[1].set_title("UnitPrice — Before vs After Mean Imputation")
sns.kdeplot(df_demo['UnitPrice'].dropna(), ax=axes[1], label='Before',
color='blue')
sns.kdeplot(df_mean['UnitPrice'], ax=axes[1], label='After',
color='red', linestyle='--')
```

```
axes[1].legend()
plt.tight_layout()
plt.show()
```

```
Mean of Quantity: 9.57
Mean of UnitPrice: 4.53

Missing values after Mean Imputation:
Quantity     0
UnitPrice    0
dtype: int64
```



Technique 3: Median Imputation

Replace missing numerical values with the **median** of the column. More robust to outliers than mean.

```
# Technique 3: Median Imputation
df_median = df_demo.copy()

median_qty = df_median['Quantity'].median()
median_price = df_median['UnitPrice'].median()

df_median['Quantity'].fillna(median_qty, inplace=True)
df_median['UnitPrice'].fillna(median_price, inplace=True)

print(f"Median of Quantity: {median_qty:.2f}")
print(f"Median of UnitPrice: {median_price:.2f}")
print(f"\nMissing values after Median Imputation:")
print(df_median[['Quantity', 'UnitPrice']].isnull().sum())

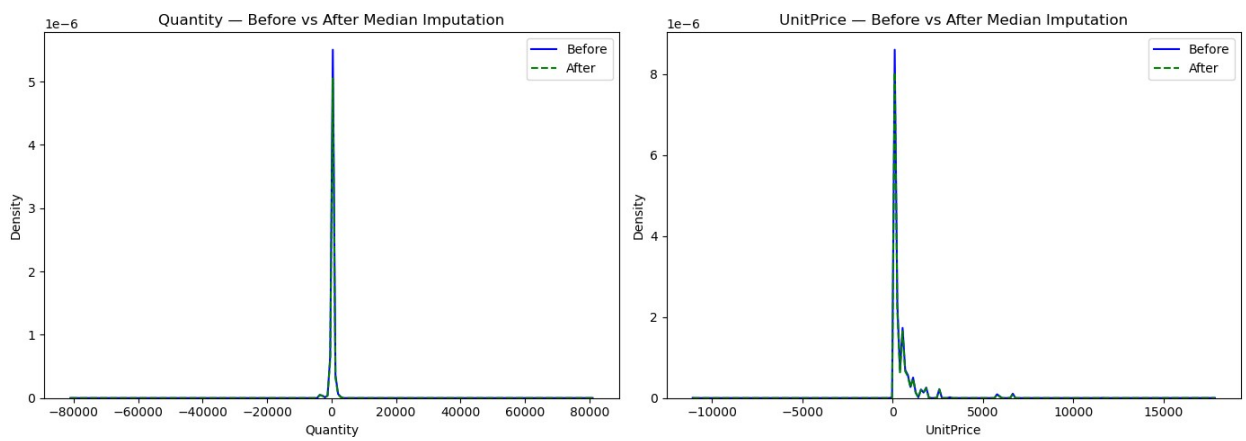
# Visualize
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
axes[0].set_title("Quantity — Before vs After Median Imputation")
sns.kdeplot(df_demo['Quantity'].dropna(), ax=axes[0], label='Before',
color='blue')
```

```
sns.kdeplot(df_median['Quantity'], ax=axes[0], label='After',
color='green', linestyle='--')
axes[0].legend()

axes[1].set_title("UnitPrice — Before vs After Median Imputation")
sns.kdeplot(df_demo['UnitPrice'].dropna(), ax=axes[1], label='Before',
color='blue')
sns.kdeplot(df_median['UnitPrice'], ax=axes[1], label='After',
color='green', linestyle='--')
axes[1].legend()
plt.tight_layout()
plt.show()
```

```
Median of Quantity: 3.00
Median of UnitPrice: 2.08

Missing values after Median Imputation:
Quantity     0
UnitPrice    0
dtype: int64
```



Technique 4: Mode Imputation

Replace missing categorical values with the **most frequent** category (mode).

```
# Technique 4: Mode Imputation (for categorical column — Description)
df_mode = df_demo.copy()

mode_desc = df_mode['Description'].mode()[0]
df_mode['Description'].fillna(mode_desc, inplace=True)

print(f"Mode of Description: '{mode_desc}'")
print(f"\nMissing values in Description after Mode Imputation:
{df_mode['Description'].isnull().sum()}")

# Bar chart of top 10 descriptions after mode imputation
```

```
top_desc = df_mode['Description'].value_counts().head(10)
fig = px.bar(x=top_desc.index, y=top_desc.values,
             labels={'x': 'Description', 'y': 'Count'},
             title='Top 10 Descriptions After Mode Imputation')
fig.update_layout(xaxis_tickangle=-45)
fig.show()
```

```
Mode of Description: 'WHITE HANGING HEART T-LIGHT HOLDER'

Missing values in Description after Mode Imputation: 0
```


```
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Technique 5: Arbitrary Value Imputation

Replace missing values with an arbitrary constant (e.g., -999 for numerical, "UNKNOWN" for categorical).

```
# Technique 5: Arbitrary Value Imputation
df_arb = df_demo.copy()

df_arb['Quantity'].fillna(-999, inplace=True)
df_arb['UnitPrice'].fillna(-999, inplace=True)
df_arb['Description'].fillna('UNKNOWN', inplace=True)

print("Arbitrary values used: -999 (numerical), 'UNKNOWN'
(categorical)")
print(f"\nMissing values after Arbitrary Imputation:")
print(df_arb[['Quantity', 'UnitPrice', 'Description']].isnull().sum())

# Show a sample of imputed rows
print("\nSample rows with arbitrary values:")
df_arb[(df_arb['Quantity'] == -999) | (df_arb['UnitPrice'] == -999) |
(df_arb['Description'] == 'UNKNOWN')].head()
```

```
Arbitrary values used: -999 (numerical), 'UNKNOWN' (categorical)

Missing values after Arbitrary Imputation:
Quantity       0
UnitPrice      0
Description    0
dtype: int64

Sample rows with arbitrary values:
```

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Technique 6: End of Tail Imputation

Replace missing values with extreme values: **mean ± 3 × std** (places imputed values at the tail of the distribution).

```
# Technique 6: End of Tail Imputation
df_tail = df_demo.copy()

# Upper tail: mean + 3*std
tail_qty = df_tail['Quantity'].mean() + 3 * df_tail['Quantity'].std()
tail_price = df_tail['UnitPrice'].mean() + 3 *
df_tail['UnitPrice'].std()

df_tail['Quantity'].fillna(tail_qty, inplace=True)
df_tail['UnitPrice'].fillna(tail_price, inplace=True)

print(f"End of Tail value for Quantity (mean + 3*std):
{tail_qty:.2f}")
print(f"End of Tail value for UnitPrice (mean + 3*std):
{tail_price:.2f}")
print(f"\nMissing values after End of Tail Imputation:")
print(df_tail[['Quantity', 'UnitPrice']].isnull().sum())

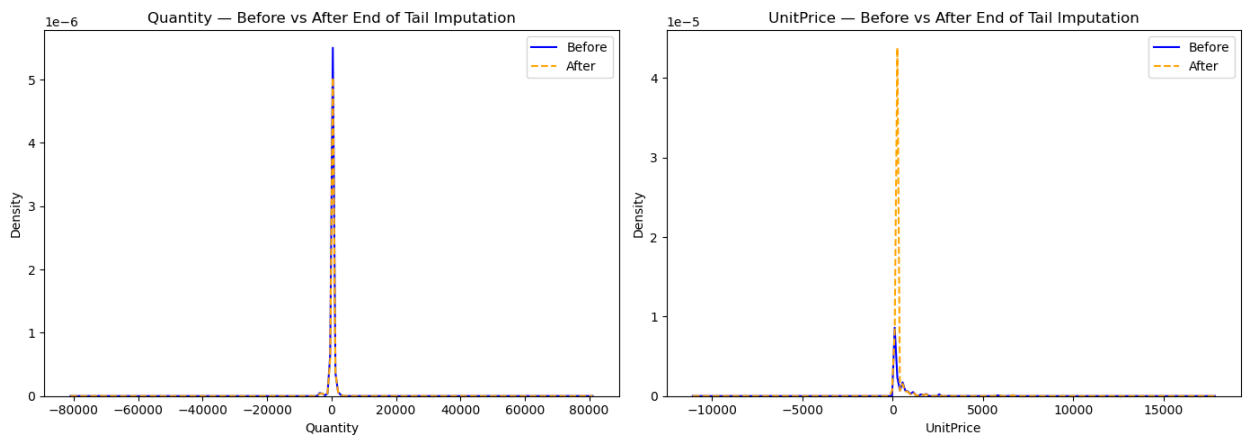
# Visualize
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
axes[0].set_title("Quantity — Before vs After End of Tail Imputation")
sns.kdeplot(df_demo['Quantity'].dropna(), ax=axes[0], label='Before',
color='blue')
sns.kdeplot(df_tail['Quantity'], ax=axes[0], label='After',
color='orange', linestyle='--')
axes[0].legend()

axes[1].set_title("UnitPrice — Before vs After End of Tail
Imputation")
sns.kdeplot(df_demo['UnitPrice'].dropna(), ax=axes[1], label='Before',
color='blue')
sns.kdeplot(df_tail['UnitPrice'], ax=axes[1], label='After',
color='orange', linestyle='--')
```

```
axes[1].legend()
plt.tight_layout()
plt.show()

End of Tail value for Quantity (mean + 3*std): 688.70
End of Tail value for UnitPrice (mean + 3*std): 240.49

Missing values after End of Tail Imputation:
Quantity     0
UnitPrice    0
dtype: int64
```



Technique 7: Random Sample Imputation

Fill missing values by randomly sampling from existing non-null values of the same column.

```
# Technique 7: Random Sample Imputation
df_rand = df_demo.copy()
np.random.seed(42)

def random_sample_impute(series):
    """Fill NaN values with random samples from non-null values."""
    missing_mask = series.isnull()
    n_missing = missing_mask.sum()
    non_null = series.dropna()
    random_samples = non_null.sample(n_missing, replace=True,
random_state=42).values
    series_copy = series.copy()
    series_copy[missing_mask] = random_samples
    return series_copy

df_rand['Quantity'] = random_sample_impute(df_rand['Quantity'])
df_rand['UnitPrice'] = random_sample_impute(df_rand['UnitPrice'])
df_rand['Description'] = random_sample_impute(df_rand['Description'])

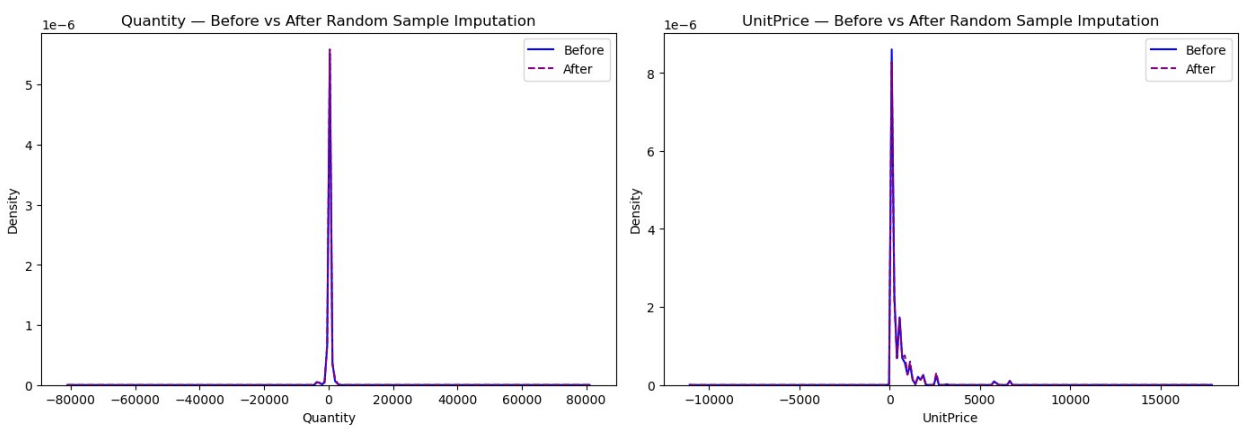
print("Missing values after Random Sample Imputation:")
```

```
print(df_rand[['Quantity', 'UnitPrice',
'Description']].isnull().sum())

# Visualize
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
axes[0].set_title("Quantity — Before vs After Random Sample
Imputation")
sns.kdeplot(df_demo['Quantity'].dropna(), ax=axes[0], label='Before',
color='blue')
sns.kdeplot(df_rand['Quantity'], ax=axes[0], label='After',
color='purple', linestyle='--')
axes[0].legend()

axes[1].set_title("UnitPrice — Before vs After Random Sample
Imputation")
sns.kdeplot(df_demo['UnitPrice'].dropna(), ax=axes[1], label='Before',
color='blue')
sns.kdeplot(df_rand['UnitPrice'], ax=axes[1], label='After',
color='purple', linestyle='--')
axes[1].legend()
plt.tight_layout()
plt.show()
```

```
Missing values after Random Sample Imputation:
Quantity       0
UnitPrice      0
Description    0
dtype: int64
```



Technique 8: Frequent Category Imputation

Replace missing categorical values with the **most frequent** category (similar to Mode for categorical data).

```
# Technique 8: Frequent Category Imputation (for Country column)
df_freq = df_demo.copy()
```

```
# Introduce some missing values in Country for demonstration
np.random.seed(10)
mask_country = np.random.choice(df_freq.index, size=int(len(df_freq) *
0.04), replace=False)
df_freq.loc[mask_country, 'Country'] = np.nan

freq_country = df_freq['Country'].mode()[0]
print(f"Most frequent Country: '{freq_country}'")
print(f"Missing Country values before:
{df_freq['Country'].isnull().sum()}")

df_freq['Country'].fillna(freq_country, inplace=True)
print(f"Missing Country values after:
{df_freq['Country'].isnull().sum()}")

# Plotly pie chart for top countries
top_countries = df_freq['Country'].value_counts().head(8)
fig = px.pie(values=top_countries.values, names=top_countries.index,
             title='Country Distribution After Frequent Category
Imputation')
fig.show()
```

```
Most frequent Country: 'United Kingdom'
Missing Country values before: 21676
Missing Country values after: 0
```


```
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Technique 9: Adding a New Category as "Missing"

Instead of imputing, treat the missing value as its own category — explicitly labeled **"Missing"**.

```
# Technique 9: Adding a New "Missing" Category
df_missing_cat = df_demo.copy()

df_missing_cat['Description'].fillna('Missing', inplace=True)
df_missing_cat['CustomerID'] = df_demo['CustomerID'].astype(str)
df_missing_cat['CustomerID'].replace('nan', 'Missing', inplace=True)

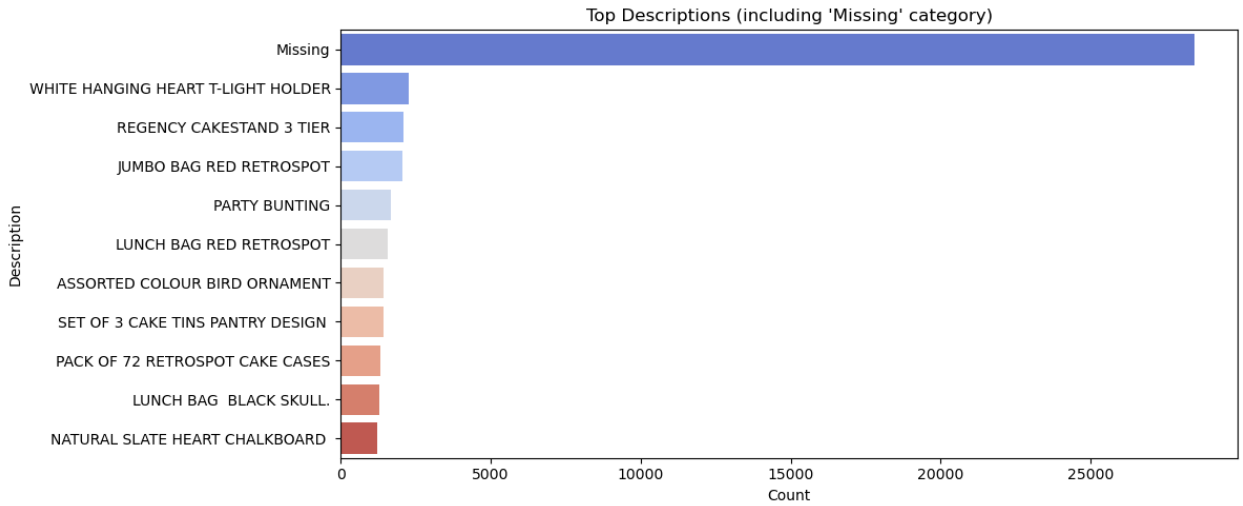
print("Missing values after adding 'Missing' category:")
print(df_missing_cat[['Description', 'CustomerID']].isnull().sum())

# Count of "Missing" category
print(f"\nRows with Description = 'Missing':
{(df_missing_cat['Description'] == 'Missing').sum()}")
print(f"Rows with CustomerID = 'Missing':
{(df_missing_cat['CustomerID'] == 'Missing').sum()}")

# Visualization
desc_counts = df_missing_cat['Description'].value_counts().head(11)
plt.figure(figsize=(12, 5))
sns.barplot(x=desc_counts.values, y=desc_counts.index,
palette='coolwarm')
plt.title("Top Descriptions (including 'Missing' category)")
plt.xlabel('Count')
plt.tight_layout()
plt.show()
```

```
Missing values after adding 'Missing' category:
Description    0
CustomerID     0
dtype: int64
```

```
Rows with Description = 'Missing': 28481
Rows with CustomerID = 'Missing': 135080
```



Technique 10: Regression Imputation

Use a **Linear Regression** model to predict missing values based on other features.

```python
# Technique 10: Regression Imputation (Predict UnitPrice from
Quantity)
df_reg = df_demo.copy()

# First, impute Quantity with median so we can use it as predictor
df_reg['Quantity'].fillna(df_reg['Quantity'].median(), inplace=True)

# Separate rows where UnitPrice is known vs missing
train = df_reg[df_reg['UnitPrice'].notna()]
test = df_reg[df_reg['UnitPrice'].isna()]

print(f"Training rows (UnitPrice known): {len(train)}")
print(f"Rows to predict (UnitPrice missing): {len(test)}")

# Fit Linear Regression: UnitPrice ~ Quantity
model = LinearRegression()
model.fit(train[['Quantity']], train['UnitPrice'])

# Predict missing UnitPrice
predicted = model.predict(test[['Quantity']])
df_reg.loc[df_reg['UnitPrice'].isna(), 'UnitPrice'] = predicted

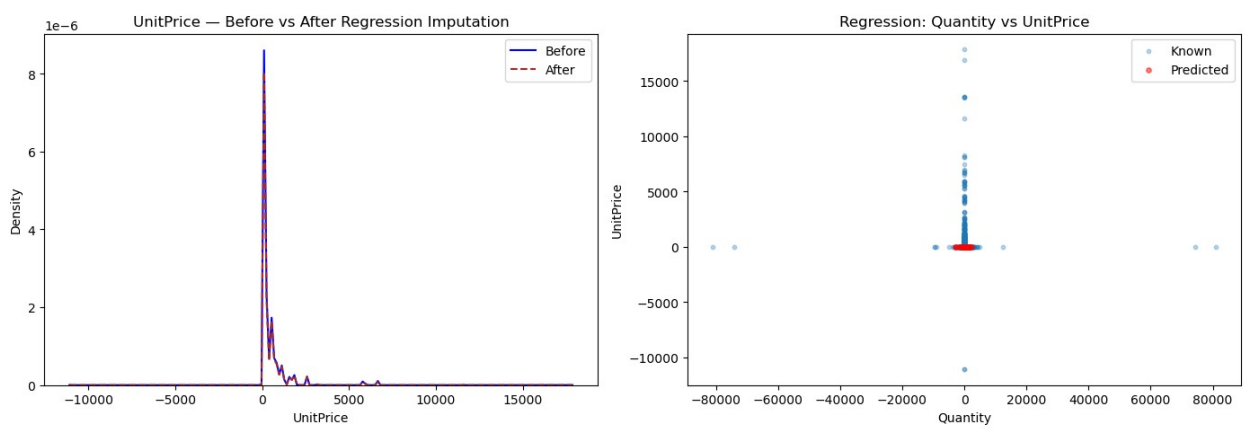
print(f"\nRegression Coefficients — Intercept: {model.intercept_:.4f},
Slope: {model.coef_[0]:.4f}")
print(f"Missing UnitPrice after Regression Imputation:
{df_reg['UnitPrice'].isnull().sum()}")
```

```
# Visualize
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
axes[0].set_title("UnitPrice — Before vs After Regression Imputation")
sns.kdeplot(df_demo['UnitPrice'].dropna(), ax=axes[0], label='Before',
color='blue')
sns.kdeplot(df_reg['UnitPrice'], ax=axes[0], label='After',
color='brown', linestyle='--')
axes[0].legend()

axes[1].scatter(train['Quantity'], train['UnitPrice'], alpha=0.3,
s=10, label='Known')
axes[1].scatter(test['Quantity'], predicted, color='red', alpha=0.5,
s=15, label='Predicted')
axes[1].set_title("Regression: Quantity vs UnitPrice")
axes[1].set_xlabel("Quantity")
axes[1].set_ylabel("UnitPrice")
axes[1].legend()
plt.tight_layout()
plt.show()
```

```
Training rows (UnitPrice known): 509395
Rows to predict (UnitPrice missing): 32514

Regression Coefficients — Intercept: 4.5362, Slope: -0.0005
Missing UnitPrice after Regression Imputation: 0
```



4. Comparison of Imputation Techniques

```
# Comparison: Effect of imputation techniques on UnitPrice statistics
original_stats = df_demo['UnitPrice'].dropna().describe()

comparison = pd.DataFrame({
    'Original (no NaN)': original_stats,
    'Mean Imputed': df_mean['UnitPrice'].describe(),
    'Median Imputed': df_median['UnitPrice'].describe(),
    'End of Tail': df_tail['UnitPrice'].describe(),
```

```
    'Random Sample': df_rand['UnitPrice'].describe(),
    'Regression': df_reg['UnitPrice'].describe()
})
print("UnitPrice Statistics — Comparison Across Techniques:\n")
comparison
```

```
UnitPrice Statistics — Comparison Across Techniques:
```


```
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```


```
# Plotly grouped bar chart — comparing mean and std across techniques
techniques = ['Original', 'Mean', 'Median', 'End of Tail', 'Random
Sample', 'Regression']
means = [
    df_demo['UnitPrice'].dropna().mean(),
    df_mean['UnitPrice'].mean(),
    df_median['UnitPrice'].mean(),
    df_tail['UnitPrice'].mean(),
    df_rand['UnitPrice'].mean(),
    df_reg['UnitPrice'].mean()
]
stds = [
    df_demo['UnitPrice'].dropna().std(),
    df_mean['UnitPrice'].std(),
    df_median['UnitPrice'].std(),
    df_tail['UnitPrice'].std(),
    df_rand['UnitPrice'].std(),
    df_reg['UnitPrice'].std()
]
```

```
fig = go.Figure(data=[
    go.Bar(name='Mean', x=techniques, y=means,
marker_color='steelblue'),
    go.Bar(name='Std Dev', x=techniques, y=stds,
marker_color='salmon')
])
fig.update_layout(
    barmode='group',
    title='UnitPrice — Mean & Std Comparison Across Imputation
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    yaxis_title='Value',
    xaxis_title='Technique'
)
fig.show()
```


```
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Conclusion

In this experiment, we applied **10 different data imputation techniques** on the Online Retail dataset:

1. **Row Deletion** — Simplest but causes significant data loss; only suitable when missing data is minimal and MCAR.
2. **Mean Imputation** — Preserves the original mean but reduces variance; not suitable when outliers are present.
3. **Median Imputation** — More robust to outliers than mean; preferred for skewed distributions.
4. **Mode Imputation** — Effective for categorical features; may overrepresent the dominant category.
5. **Arbitrary Value Imputation** — Clearly flags missing data but can distort distributions if values are unrealistic.
6. **End of Tail Imputation** — Pushes imputed values to extremes; useful when missingness itself is informative.
7. **Random Sample Imputation** — Best preserves the original distribution; introduces randomness.
8. **Frequent Category Imputation** — Simple for categorical data; biases toward the most common class.
9. **"Missing" Category** — Retains missingness as information; increases feature cardinality.
10. **Regression Imputation** — Leverages feature relationships; assumes linearity and can overfit.

Key Takeaway: No single imputation method is universally best — the choice depends on the data distribution, the mechanism of missingness (MCAR, MAR, MNAR), and the downstream analysis requirements. Visualizing missing patterns before imputation is crucial for making informed decisions.